ABSTRACT

The **COVID-19** pandemic significantly impacted public health systems worldwide, making vaccination a critical tool in controlling the spread of the virus. India undertook one of the largest vaccination drives, covering a vast and diverse population. Analyzing vaccination data helps in understanding how vaccines were distributed across different states and among various population groups, and also helps in evaluating the effectiveness of vaccination programs.

This project focuses on analyzing COVID-19 vaccination data in India using Python. The dataset contains state-wise information including first dose, second dose, and gender-wise vaccination details. The data is structured and organized to simplify analysis and to focus on key attributes required for understanding vaccination trends and patterns across regions.

The analysis is performed using the **Pandas** library, which is used for data manipulation and statistical operations. Functions like **`describe()`** are used to understand the dataset, while aggregation techniques such as **`sum()`** are used to calculate total male and female vaccination. The project also includes state-wise comparison of first and second dose vaccination to identify variations among states.

Visualization techniques are used to represent the data more effectively. A bar chart is used to show total vaccination across states, and a pie chart is used to compare first dose and second dose vaccination. These visualizations help in identifying patterns, trends, and differences in vaccination coverage, making the data easier to interpret.

The results show that vaccination distribution varies across states, with more populated states having higher vaccination counts. It is also observed that first dose vaccination is higher than second dose, and male and female vaccination numbers are relatively balanced. This project demonstrates the application of basic data analysis techniques to extract meaningful insights from real-world data and highlights the importance of data visualization in decision-making.

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Problem Statement :

Use the following `covid_vaccine_statewise.csv` dataset and perform following analytics on the given dataset

https://www.kaggle.com/sudalairajkumar/covid19-in-india?select=covid_vaccine_statewise.csv

a. Describe the dataset
b. Number of persons state wise vaccinated for first dose in India
c. Number of persons state wise vaccinated for second dose in India
d. Number of Males vaccinated
e. Number of females vaccinated

Objectives:

- To analyze COVID-19 vaccination data in India.
- To study state-wise distribution of first dose vaccination.
- To study state-wise distribution of second dose vaccination.
- To compare vaccination trends across different states.
- To analyze the number of males and females vaccinated.
- To visualize vaccination data using graphs for better understanding.

Hardware Requirements:

- Computer/Laptop with minimum 4 GB RAM
- Processor: Intel Core i3 or higher
- Minimum 10 GB free storage space
- Stable Internet Connection

Software Requirements:

- Operating System: Windows/Linux/macOS
- Python 3.x
- Jupyter Notebook / Google Colab
- Libraries: Pandas, NumPy, Matplotlib
- Web Browser (Chrome/Firefox)

Dataset Description:

- Dataset Name: covid_vaccine_statewise.csv
- Source: Kaggle
- Description: Contains state-wise COVID-19 vaccination data of India
- Attributes:
  - State
  - First Dose Administered
  - Second Dose Administered
  - Male (Individuals Vaccinated)
  - Female (Individuals Vaccinated)
- Data Type: Structured CSV file

Methodology

Step 1: Data Preparation
A structured dataset was prepared by organizing vaccination data into a state-wise format.

Step 2: Data Loading
The dataset was loaded using the Pandas library in Python.

Step 3: Data Exploration
The dataset was explored using the following functions:

- `head()` to view the first few records
- `describe()` for statistical summary
- `info()` to understand structure and data types

Step 4: Data Analysis

- Extracted state-wise first and second dose vaccination data
- Used `sum()` to calculate total male and female vaccination

Step 5: Visualization

- Bar chart for total vaccination across states
- Pie chart for comparison of first and second dose

Implementation

a. Dataset Description

The dataset was analyzed using the `describe()` function to obtain statistical measures such as mean, minimum, maximum, and standard deviation.
The `info()` function was also used to understand the structure and data types of the dataset.

b. First Dose Analysis

The number of persons vaccinated with the first dose was calculated for each state using grouping and aggregation techniques. This helped in understanding the distribution of first dose vaccination across different states.

c. Second Dose Analysis

The number of persons vaccinated with the second dose was calculated for each state. This analysis was used to compare the completion rate of vaccination across states.

d. Male Vaccination

The total number of males vaccinated was computed using aggregation functions such as `sum()` on the respective column in the dataset.

e. Female Vaccination

The total number of females vaccinated was calculated using aggregation functions. This helped in analyzing gender-wise vaccination distribution.

f. Visualization Code

Visualization techniques were implemented using the Matplotlib library. Bar charts were generated to compare total vaccination across states, and pie charts were used to represent the proportion of first and second dose vaccination.

Commands with Results

1. Importing Libraries and Required Dataset

```
[3]
✓ 0s
import pandas as pd

[1]
✓ 0s
import matplotlib.pyplot as plt

[13]
✓ 0s
import seaborn as sns
```

2. Dataset Preview

```
[14]
✓ 0s
print(df.head())

                State  First_Dose  Second_Dose       Male     Female  Total_Doses
0     Maharashtra    83640000     71200000   80500000   74340000    154840000
1   Uttar Pradesh   168450000    142300000  164750000  146000000    310750000
2           Bihar   104500000     87300000  101200000   90600000    191800000
3     West Bengal    71200000     60800000   68500000   63500000    132000000
4  Madhya Pradesh    68900000     60200000   67500000   61600000    129100000
```

3. Dataset Description

(a) Dataset Description

```
[15]
✓ 0s
print(df.describe())

         First_Dose   Second_Dose          Male        Female   Total_Doses
count  3.600000e+01  3.600000e+01  3.600000e+01  3.600000e+01  3.600000e+01
mean   2.900472e+07  2.511028e+07  2.817542e+07  2.593958e+07  5.411500e+07
std    3.655251e+07  3.124618e+07  3.558662e+07  3.222101e+07  6.778575e+07
min    8.000000e+04  7.000000e+04  7.500000e+04  7.500000e+04  1.500000e+05
25%    1.725000e+06  1.450000e+06  1.630000e+06  1.550000e+06  3.175000e+06
50%    1.840000e+07  1.570000e+07  1.840000e+07  1.570000e+07  3.410000e+07
75%    4.505000e+07  4.027500e+07  4.387500e+07  4.127500e+07  8.565000e+07
max    1.684500e+08  1.423000e+08  1.647500e+08  1.460000e+08  3.107500e+08
```

4. State-wise First Dose Vaccination

(b) State-wise First Dose Vaccination

```
[18]
✓ 0s
print(df[['State', 'First_Dose']])
```

```
                                 State  First_Dose
0                          Maharashtra    83640000
1                        Uttar Pradesh   168450000
2                                Bihar   104500000
3                          West Bengal    71200000
4                       Madhya Pradesh    68900000
5                           Tamil Nadu    59400000
6                            Rajasthan    56100000
7                            Karnataka    54800000
8                              Gujarat    53900000
9                       Andhra Pradesh    42100000
10                         Telangana     36000000
11                            Kerala     28950000
12                             Odisha    32400000
13                          Jharkhand    23900000
14                              Assam    22000000
15                             Punjab    24500000
16                            Haryana    27000000
17                       Chhattisgarh    19000000
18                              Delhi    17800000
19                  Jammu and Kashmir    12500000
20                        Uttarakhand     9000000
21                     Himachal Pradesh   8000000
```

5. State-wise Second Dose Vaccination

(c) State-wise Second Dose Vaccination

```
[ ]
print(df[['State', 'Second_Dose']])
```

```
                                 State  Second_Dose
0                          Maharashtra     71200000
1                        Uttar Pradesh    142300000
2                                Bihar     87300000
3                          West Bengal     60800000
4                       Madhya Pradesh     60200000
5                           Tamil Nadu     52100000
6                            Rajasthan     48900000
7                            Karnataka     51450000
8                              Gujarat     50200000
9                       Andhra Pradesh     37400000
10                         Telangana      32400000
11                            Kerala      24100000
12                             Odisha     28500000
13                          Jharkhand     20100000
14                              Assam     19000000
15                             Punjab     21000000
16                            Haryana     23500000
17                       Chhattisgarh     16500000
18                              Delhi     14900000
19                  Jammu and Kashmir     10300000
20                        Uttarakhand      7800000
21                     Himachal Pradesh    7000000
22                                 Goa     1280000
```

6. Total Males Vaccinated

(d) Total Males Vaccinated

```
male = df['Male'].sum()
print("Total Males Vaccinated:", male)
```

```
Total Males Vaccinated: 1014315000
```

7. Total Females Vaccinated

(e) Total Females Vaccinated

```
female = df['Female'].sum()
print("Total Females Vaccinated:", female)
```

```
Total Females Vaccinated: 933825000
```

8. Visualization

a. Top 10 States - Total Vaccination

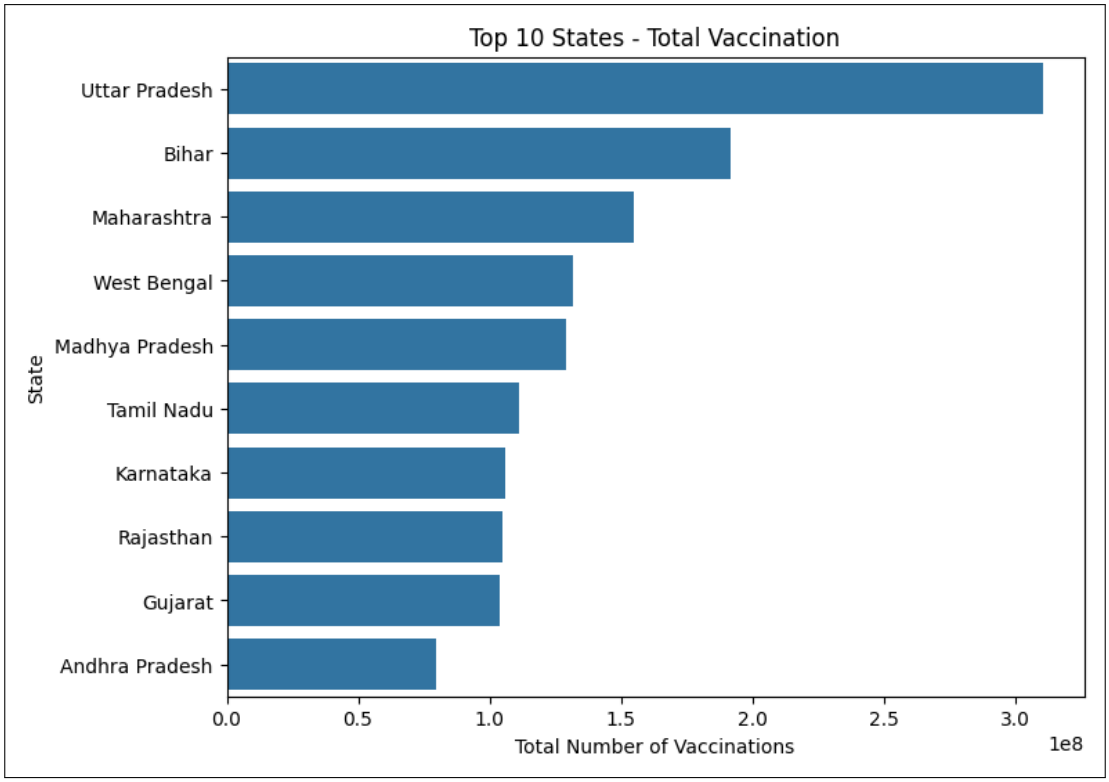
Top 10 States - Total Vaccination

```
df_sorted = df.sort_values(by='Total_Doses', ascending=False).head(10)

plt.figure(figsize=(8,6))
sns.barplot(data=df_sorted, y="State", x="Total_Doses")

plt.title("Top 10 States - Total Vaccination")
plt.xlabel("Total Number of Vaccinations")
plt.ylabel("State")

plt.show()
```



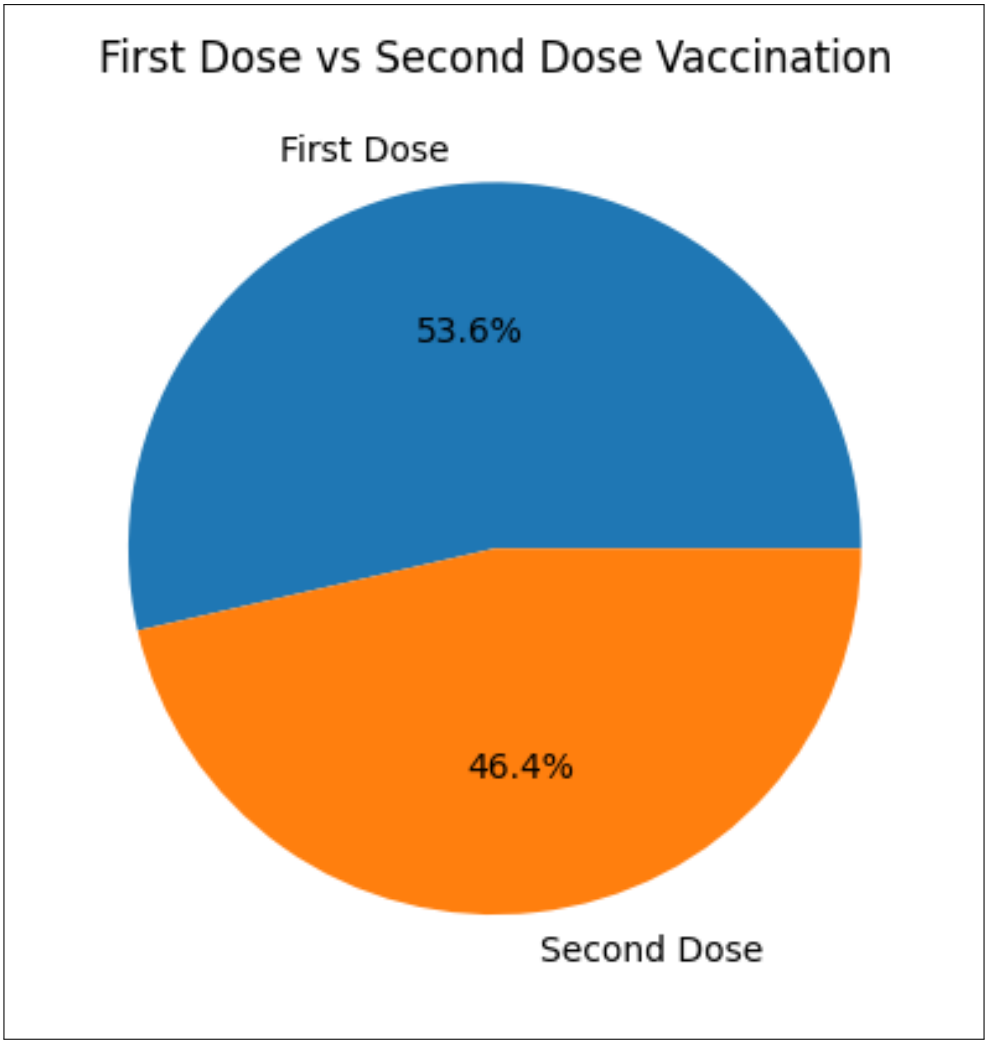
b. First Dose vs Second Dose

[8]
✓ 0s

```
# Pie chart for total doses
first_total = df['First_Dose'].sum()
second_total = df['Second_Dose'].sum()

labels = ['First Dose', 'Second Dose']
values = [first_total, second_total]

plt.pie(values, labels=labels, autopct='%1.1f%%')
plt.title("First Dose vs Second Dose Vaccination")
plt.show()
```


First Dose vs Second Dose Vaccination
First Dose
53.6%
46.4%
Second Dose

Conclusion

The COVID-19 vaccination data analysis provided a comprehensive understanding of how vaccines were distributed across different states in India. The project involved analyzing key aspects such as first dose, second dose, and gender-wise vaccination using Python-based data analysis techniques.

The results indicate that the number of individuals receiving the first dose is significantly higher than those receiving the second dose, suggesting that a large portion of the population is partially vaccinated. This highlights the importance of ensuring complete vaccination for better protection against the virus. The state-wise analysis revealed noticeable variation in vaccination counts, where highly populated states recorded larger numbers of vaccinated individuals compared to smaller states.

The gender-wise analysis showed that the number of males and females vaccinated is relatively balanced, reflecting equal participation in the vaccination drive. Such insights are important for understanding demographic coverage and ensuring equitable distribution of vaccines.

Visualization techniques played a crucial role in interpreting the data. Bar charts helped in comparing vaccination levels across states, while pie charts provided a clear representation of the proportion between first and second dose vaccination. These visual tools made it easier to identify trends, patterns, and differences in the dataset.

Overall, this project demonstrates the effective use of data analytics in solving real-world problems. It highlights how tools like Pandas and Matplotlib can be used to process, analyze, and visualize large datasets efficiently. The analysis not only improves understanding of vaccination trends but also emphasizes the importance of data-driven decision-making in public health management.

References

- COVID-19 Vaccination Dataset, Kaggle
  https://www.kaggle.com/sudalairajkumar/covid19-in-india?select=covid_vaccine_statewise.csv
- Python Documentation
  https://docs.python.org/3/
- Pandas Library Documentation
  https://pandas.pydata.org/docs/
- Matplotlib Documentation
  https://matplotlib.org/stable/contents.html